DISORDERS OF DNA REPAIR

Xeroderma Pigmentosum

Xeroderma pigmentosum (XP) serves as the prototype heritable disease with increased sensitivity to cellular injury. It is an autosomal recessive disorder characterized by extreme sun sensitivity, photophobia, early onset of freckling, and a high incidence of skin cancers on sun-exposed areas. Patients exhibit cellular hypersensitivity to ultraviolet (UV) radiation and certain DNA-damaging chemicals, due to defects in DNA repair.

Eight clinical phenotypes of DNA repair diseases have been recognized (e.g., XP, Cockayne syndrome, trichothiodystrophy). Mutations in 12 genes are associated with these phenotypes, with significant phenotypic overlap. Notably, mutations in a single gene can lead to multiple distinct clinical syndromes, and different genes can cause similar phenotypes.

For example, mutations in the **XPD (ERCC2)** gene are linked to: - Xeroderma pigmentosum (XP) - XP with neurologic abnormalities - XP-Cockayne syndrome complex - Trichothiodystrophy (TTD) - XP-trichothiodystrophy - COFS syndrome (cerebro-oculo-facio-skeletal syndrome)

The official gene names are listed in parentheses; complementation groups are denoted in uppercase (e.g., XPD).

Nucleotide Excision Repair and Related Disorders

The **nucleotide excision repair (NER)** pathway repairs bulky DNA lesions, such as UV-induced photoproducts. NER is impaired in: - **Xeroderma pigmentosum (XP)** - **Cockayne syndrome (CS)** - **Trichothiodystrophy (TTD)**

Cells from patients with these disorders show increased sensitivity to UV light, with higher rates of cell death and mutation. However, only XP patients have a markedly increased risk of skin cancer. The reason for this differential cancer risk remains unclear.

Some proteins involved in these conditions have dual roles in **DNA repair** and **transcription**, particularly as components of the basal transcription factor **TFIIH**. For instance: - **XPB (ERCC3)**: DNA helicase, unwinds DNA in the 3′ to 5′ direction - **XPD (ERCC2)**: DNA helicase, unwinds DNA in the 5′ to 3′ direction

Mutations affecting different functional domains of these proteins may lead to distinct clinical phenotypes, contributing to overlap syndromes such as XP/CS or XP/TTD.

XP Variant: Defect in DNA Damage Tolerance

The **XP variant** phenotype is clinically indistinguishable from classical XP but results from a defect in **translesional DNA synthesis**—a DNA damage tolerance mechanism—rather than NER. XP-variant cells cannot efficiently bypass DNA lesions during replication, leading to replication fork stalling and genomic instability.

Mismatch Repair and Cancer Syndromes

The **DNA mismatch repair (MMR)** pathway is defective in: - **Hereditary nonpolyposis colorectal cancer (HNPCC)**, also known as **Lynch syndrome** - **Muir-Torre syndrome**, a variant of Lynch syndrome with cutaneous manifestations such as sebaceous adenomas and carcinomas

These conditions are associated with microsatellite instability and increased risk of colorectal, endometrial, and other cancers.

DNA Helicases and Genomic Stability

Helicases are essential for DNA unwinding during replication, transcription, repair, and recombination. The **RecQ family** of DNA helicases is highly conserved from bacteria to humans. Three human RecQ helicases are linked to genetic disorders: - **BLM** (mutated in **Bloom syndrome, BS**): chromosomal instability, cancer predisposition - **WRN** (mutated in **Werner syndrome, WS**): premature aging, cancer risk - **RECQL4** (mutated in **Rothmund-Thomson syndrome, RTS**): skeletal abnormalities, skin changes, cancer predisposition

All three are associated with chromosomal instability and increased cancer risk.

BASC and Genome Surveillance

The **BASC (BRCA1-associated genome surveillance complex)** is a nuclear multi-enzyme complex centered on **BRCA1**. It includes key DNA damage response proteins such as: - **ATM** - **ATR** - **MRE11-RAD50-NBS1 (MRN) complex** - **BRCA1** - **BLM** - **FANCD2** - **MLH1**

These proteins physically interact to coordinate DNA damage detection, repair, and cell cycle control. Defects in any of these components can lead to genome instability and are associated with disorders including: - **Ataxia telangiectasia** - **Seckel syndrome** - **Nijmegen breakage syndrome** - **Hereditary breast and ovarian cancer (BRCA1/2)** - **Bloom syndrome (BS)** - **Fanconi anemia (FA)** - **Hereditary colon cancer**

Bloom syndrome and Fanconi anemia are discussed in detail in the following sections.